

VigilVoice

Voice evidence.

Verified actions.

A voice-risk detector connected to a guarded approval workflow.

Real pretrained audio inference · Separate verification · Simulated transfers · Auditable decisions

Teacher demonstration

No money moves

A convincing voice should not be enough to authorize a sensitive action.

The risk

A synthetic recording can sound persuasive. Poor recordings also make audio assessment unreliable.

The operator needs a clear decision process when evidence is risky or missing.

Our prototype

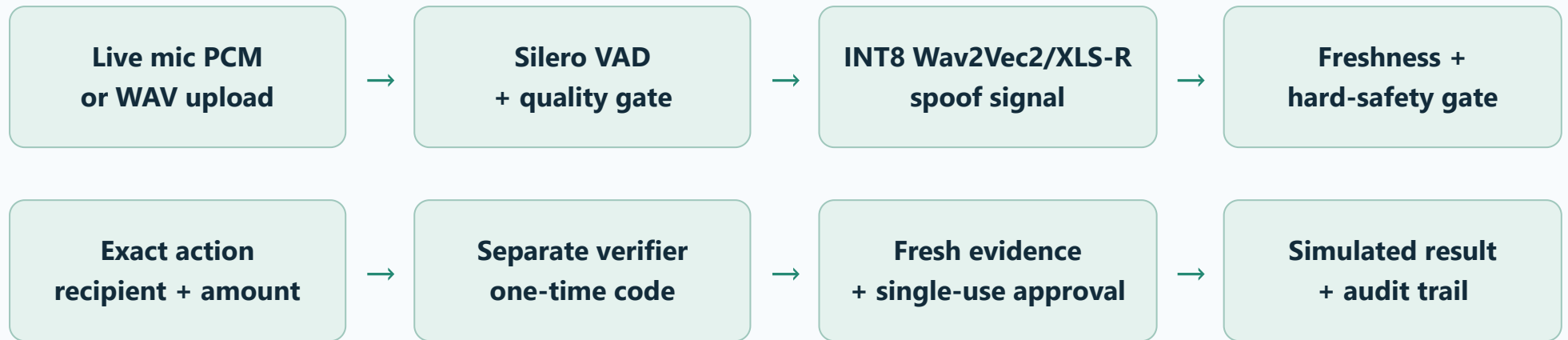
Assess speech quality and spoof signals, then require verification of the exact recipient and amount.

Block completion when evidence is unsafe, missing, stale, or verification fails.

The audio score is a risk signal. It does not establish identity, intent, or a calibrated probability of fraud.

WHAT IS IMPLEMENTED

One traceable path from speech to decision.



FastAPI · PostgreSQL · ONNX Runtime on CPU · Native browser UI · Docker

Pretrained third-party models run locally. The browser streams sequenced 200 ms PCM16 frames; record-then-analyze WAV remains a fallback. Optional Jev shadow sees metadata only and cannot authorize an action.

Three cases the audience can follow.

Input	What to demonstrate	Expected workflow
Live microphone	Start detection; speak clearly; watch risk update	Show the actual result, including HIGH or quality rejection
Bundled genuine WAV	Analyze → prepare transfer → verifier reviews details → code → complete	LOW/ELEVATED still requires action-bound verification
Bundled synthetic WAV	New transfer → analyze → prepare action	If HIGH, action blocks; report any miss honestly

A second tab simulates delivery; it is not an independent authentication factor. Every transfer is simulated. Keep the verifier key private.

Rehearsal: file evidence 120s · code 60s · approval 60s · maximum 3 incorrect code attempts. Expired/blocked action? Start a new transfer.

Working controls, exploratory ML results.

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Isolated API check groups passed

Action binding, OTP limits, one-time completion, HIGH retention, stream failure and audit rollback.

Real local models; not production security certification.

40

Historical English clips

20 validation + 20 previously viewed “test” clips; this is regression evidence, not an untouched final test.

Speaker and model-training overlap remain unverified.

Historical 20-clip partition	Deployed HIGH 0.75	Offline candidate 0.61
HIGH synthetic recall	7 / 9 scorable	9 / 9 scorable
Genuine falsely blocked as HIGH	1 / 9 scorable	2 / 9 scorable
Quality rejections (excluded above)	2 / 20 total	2 / 20 total

The candidate is not deployed. On this small English corpus, catching more synthetic clips also blocked more genuine speech. Warm 10-second WAV p95 was 5.26s in an isolated repeated-clip runtime check.

Sources: Stage 0, V02 and V03 receipts in docs/validation. No four-language or production accuracy claim.

What we can—and cannot—claim today.

Implemented and demonstrated	Still pending
Continuous local microphone stream and WAV fallback	Representative Hindi, Hinglish, Marathi and channel validation
Real pretrained VAD and spoof inference	Defensible model/threshold freeze and untouched final test
Hard safety gate, local verifier and simulated action audit	Trusted delivery and real payment integration
Bounded single-stream test and isolated failure checks	Final-build offline and separate-verifier rehearsal
Optional metadata-only Jev shadow fallback	Live-key shadow evaluation; advisory is not approved

Present this as an engineering prototype. Do not claim an accuracy percentage for real-world use or that all SIH PDF promises are complete.

NEXT DEVELOPMENT STEPS

Make the evidence stronger, then expand deployment.

1. Acquire consented, source-linked Indian-language, replay, phone and diverse synthetic speech.
2. Evaluate source/speaker-disjoint validation, freeze model and policy, then run one untouched final test; defer replacement if data is unavailable.
3. Complete the independent-verifier and offline presentation rehearsals.
4. Before any real deployment, add trusted delivery, accounts, HTTPS and an audio-retention policy.

Current deliverable: a local prevention-first demonstration with test receipts and explicit detector limits.